Bagging and Random forest

Made with Goodnotes

Bias Variance

If training data has errors so high bias

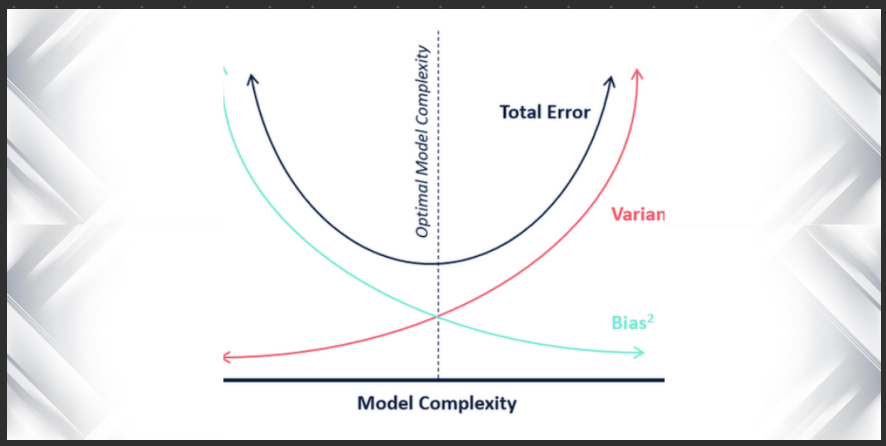
If testing data has errors so high variance

high bias & high variance

Underfitting

Low Bias & high variance

overfitting



Dataset – 1000 Data points

(max_depth = None)

m1 → Decision tree

Testing Accuracy
↓
62-64%

Training Accuracy → overfit
96-97%

k = 10

1000

m1 M2 M3 M4 ----- n

Bootstrap
1000

0 1 1 1

[1, 2, 3, 4, 5]

[1, 1, 3, 2, 2]

Majority voting → 1

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Low Bias & Low variance

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